

Validity Analysis: NLLB

Target context: Amhara Region Agricultural and Microfinance Translation Cohort

Overall risk: **HIGH**

Deployment Context

Use case: Document-level translation of agricultural input subsidy notices, land-use policy updates, and credit-scheme terms from federal and regional bureaus into Amharic

Target population: Farmer cooperative leaders and rural microfinance clients (e.g., ACSI, ADCSI borrowers) in Amhara interpreting government and lender document

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology $\triangle$	2	Significant gaps	high
Input Content $\triangle$	2	Significant gaps	high
Input Form $\checkmark$	4	Minor gaps	high
Output Ontology $\triangle$	2	Significant gaps	high
Output Content $\triangle$	2	Significant gaps	high
Output Form $\checkmark$	4	Minor gaps	high
Average	2.7		

$\triangle$ = highest concern $\checkmark$ = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

Flores-200/NLLB provides credible language- and script-level support for Amharic — Ge'ez script is included with a dedicated SentencePiece vocabulary [Q382], and the benchmark's text-only output form aligns with the deployment's accepted format [elicitation A4]. These are real strengths for this text-only Amharic deployment. However, the four HIGH-priority dimensions (Input Ontology, Input Content, Output Ontology, Output Content) show substantial validity gaps. The benchmark's task taxonomy excludes agricultural, land-tenure, and microfinance genres entirely [Q95, Q166-Q169]; its content is exclusively Wikimedia-derived [Q95, Q151] with the authors themselves acknowledging cultural-coverage limitations [Q151]; its output ontology offers only general translation-quality scalars [Q455, Q648] that cannot detect the high-consequence error types

(numerical-threshold alteration, conditional-logic distortion, terminology drift) that matter for binding administrative documents; and its annotator workforce is professional generalist translators [Q97-Q103], not the federal bureau translators with Ethiopian government-domain expertise that the deployer identifies as the validation authority [elicitation A3]. Independent 2025 research [WEB-10] empirically confirms that Flores+ scores fail to predict out-of-domain translation quality and that chrF++/BLEU can reward superficial named-entity copying. Web search found no Ethiopian agricultural/administrative MT benchmark exists [WEB-3, WEB-4] and no public ACSI/ADCSI Amharic terminology resource [WEB-15, WEB-16], meaning the gaps are real and unaddressed by the broader research community. Aggregate NLLB scores on Flores-200 should be treated as a coarse general-domain capability signal for Amharic, not as evidence of fitness for binding subsidy, land-policy, or credit documents.

ID	Evidence
Q95	“As a significant extension of Flores-101, Flores-200 consists of 3001 sentences sampled from English-language Wikimedia projects for 204 total languages. Approximately one third of sentences are collected from each of these sources: Wikinews, Wikijunior, and Wikivoyage. The content is professionally translated into 200+ languages to create Flores-200. As we translate the same set of content into all languages, Flores-200 is a many-to-many multilingual benchmark.” (p.19)
Q97	“Translators are required to have at least two to three years of translation experience in the relevant language pair if they have an academic degree in translation or linguistics and three to five years of translation experience if they do not have any relevant academic qualification.” (p.20)
Q98	“Translators also undergo a translation test every 18 months to assess their translation quality.” (p.20)
Q99	“Flores-200 reviewers are also required to be native speakers of the target language.” (p.20)
Q100	“Reviewers typically have a translation degree, at least five years of experience working as a translator, translation review experience, and where possible are accredited by a relevant translation board.” (p.20)
Q101	“We note that these are stringent standards, and extensions of Flores-200 to even more low-resource languages in the future may be difficult.” (p.20)
Q102	“Already for many languages, finding a reviewer that meets the criteria above is very challenging.” (p.20)
Q103	“In these cases, we modified the qualification process to accept applications from reviewers with more general language degrees such as Linguistics or African Language Studies, or no degree provided they have had extensive commercial translation experience (e.g. >10 years).” (p.20)
Q151	“We note that like Flores-200, NLLB-Seed's source data is English-centric and sampled from English Wikipedia. This has an important effect: the content reflects what Wikipedia editors find is relevant for English Wikipedia, and likely does not cover diverse content from different cultures.” (p.24)
Q166	“We translate the English side of the WMT21 English-German development set, containing a sample of newspapers from 2020 (Akhbardeh et al., 2021).” (p.25)
Q167	“We translate text extracted from a series of scripted English-language talks covering a variety of topics.” (p.25)
Q168	“We extract 3,000 utterances from the multi-session chat dataset of Xu et al. (2022), which contains on average 23 words per turn.” (p.25)
Q169	“We translated one World Health Organisation report (Donaldson and Rutter, 2017) and combined it with sentences translated from the English portion of the TAUS Corona Crisis Report.” (p.25)
Q382	“We then trained an encoder for Amharic and Tigrinya only, paired with English as in all our experiments, and a specific 8k SentencePiece vocabulary to better support the Ge'ez script.” (p.45)
Q455	“We use the chrF++ metric to compare the model performance (see Section 7.1).” (p.53)
Q648	“In this work, we primarily evaluate using chrF++ using the settings from sacrebleu. However, when comparing to other published works, we utilize BLEU and spBLEU where appropriate.” (p.73)
WEB-3	No Ethiopian agricultural or administrative-domain Amharic benchmark exists (https://arxiv.org/abs/2403.13737)
WEB-4	EthioMT parallel corpora (15 Ethiopian languages) drawn from religious and general domains, not administrative/financial (https://arxiv.org/html/2403.19365v1)
WEB-10	2025 study: mock translations merely reproducing named entities achieved non-trivial BLEU and ChrF++ scores, confirming metrics can reward superficial overlap rather than translational adequacy (https://arxiv.org/html/2508.20511v1)
WEB-15	ACSI official website — no Amharic financial terminology guide publicly available (http://www.acsi.org.et/)
WEB-16	FinDev Gateway ACSI profile — no terminology resource (https://www.findevgateway.org/case-study/2004/01/amhara-credit-and-saving-institution-acsi-institutional-profile-current-status)

Practical Guidance

What This Benchmark Measures

NLLB/Flores-200 measures general-domain sentence-level translation quality for Amharic prose, anchored in English-Wikimedia-derived content [Q95] and scored with chrF++/spBLEU/calibrated XSTS [Q455, Q648, Q654]. The strongest dimensions for this deployment are Input Form and Output Form: the benchmark validly assesses whether a model can produce coherent Amharic text in Ge'ez script for general-purpose prose, which is a necessary precondition for the deployment.

Construct Depth

The benchmark probes general translation capability shallowly across many directions rather than deeply for any one domain. For Amharic specifically, evaluation is limited to 3,001 Wikimedia-derived sentences [Q95] scored with sentence-level metrics [Q336] that 2025 evidence shows can reward superficial overlap [WEB-10]. There is no evidence depth on: (a) terminology consistency across a multi-clause document, (b) numerical-threshold fidelity, (c) conditional-logic preservation, (d) acceptance by Ethiopian federal bureau translators, or (e) any agricultural/financial/bureaucratic register. The benchmark's evidence stops well short of what is needed for binding administrative document deployment.

What Else You Need

To achieve a complete assessment for this deployment, the user should commission: (1) a domain-specific Amharic evaluation set covering agricultural input subsidies, land-tenure proclamations, and ACSI/ADCSI credit documents (Input Ontology + Input Content gaps); (2) a domain rubric with rules for numerical-fidelity, terminology-consistency, and conditional-logic preservation, since chrF++ cannot detect these error types (Output Ontology gap); (3) a federal-bureau-translator annotator pool to produce reference translations and acceptability ratings for representative documents (Output Content gap); (4) document-level coherence and terminology-consistency scoring on multi-clause samples (Output Form residual gap); and (5) a targeted Amharic Flores-200 audit analogous to those done for Hausa/Sepedi/Xitsonga/isiZulu [WEB-14], since no such audit has been published.

ID	Evidence
Q95	"As a significant extension of Flores-101, Flores-200 consists of 3001 sentences sampled from English-language Wikimedia projects for 204 total languages. Approximately one third of sentences are collected from each of these sources: Wikinews, Wikijunior, and Wikivoyage. The content is professionally translated into 200+ languages to create Flores-200. As we translate the same set of content into all languages, Flores-200 is a many-to-many multilingual benchmark." (p.19)
Q336	"We look for best performance on the Flores-200 development set, and report detokenized BLEU on Flores-200 devtest." (p.41)
Q455	"We use the chrF++ metric to compare the model performance (see Section 7.1)." (p.53)
Q648	"In this work, we primarily evaluate using chrF++ using the settings from sacrebleu. However, when comparing to other published works, we utilize BLEU and spBLEU where appropriate." (p.73)
Q654	"XSTS rates each source sentence and its machine translation on a five-point scale, where 1 is the lowest score and 5 is the highest score." (p.74)
WEB-10	2025 study: mock translations merely reproducing named entities achieved non-trivial BLEU and ChrF++ scores, confirming metrics can reward superficial overlap rather than translational adequacy (https://arxiv.org/html/2508.20511v1)
WEB-14	Post-publication FLORES-200 corrections exist for Hausa, Sepedi, Xitsonga, isiZulu — but not Amharic, leaving Amharic reference unaudited (https://www.emergentmind.com/topics/flores-200-benchmark-dataset)

Input Ontology (IO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: NLLB | Context: Amhara Region Agricultural and Microfinance Translation Cohort

Input Ontology definition: Whether the benchmark's test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The benchmark's task taxonomy is organized along resource level (high/low/very low) and a narrow set of domains (news, scripted talks, chat, WHO health) tested only in NLLB-MD for six language directions [\[Q159, Q166-Q169\]](#), none of which include Amharic. The deployment's core document genres — agricultural input subsidy notices, land-use policy updates, and credit-scheme terms — fall entirely outside the evaluation taxonomy. The paper itself frames the generalization question as 'How well do models generalize to non-Wikimedia domains?' [\[Q159\]](#) but does not extend domain testing to administrative, agricultural, or financial genres. Independent 2025 evidence confirms Flores+ scores do not reliably predict out-of-domain translation quality [\[WEB-10\]](#). Given the HIGH priority assigned to this dimension by the elicitation, the absence of any agricultural/bureaucratic/financial category in the benchmark taxonomy is a significant content-validity concern.

ID	Evidence
Q159	"More specifically, we want to answer the following two questions: (1) How well do models generalize to non-Wikimedia domains? (2) Does fine-tuning on high quality in-domain parallel text lead to good performance?" (p.24)
Q166	"We translate the English side of the WMT21 English-German development set, containing a sample of newspapers from 2020 (Akhbardeh et al., 2021)." (p.25)
Q167	"We translate text extracted from a series of scripted English-language talks covering a variety of topics." (p.25)
Q168	"We extract 3,000 utterances from the multi-session chat dataset of Xu et al. (2022), which contains on average 23 words per turn." (p.25)
Q169	"We translated one World Health Organisation report (Donaldson and Rutter, 2017) and combined it with sentences translated from the English portion of the TAUS Corona Crisis Report." (p.25)
WEB-10	2025 study: mock translations merely reproducing named entities achieved non-trivial BLEU and ChrF++ scores, confirming metrics can reward superficial overlap rather than translational adequacy (https://arxiv.org/html/2508.20511v1)

Strengths

- The benchmark explicitly anticipates the domain-generalization question and creates NLLB-MD to probe it [\[Q159\]](#), providing a documented framework (if not coverage) for thinking about non-Wikimedia domains.
- Resource-level stratification (high/low/very-low) [\[Q419, Q420\]](#) gives users a coarse signal of expected difficulty for Amharic, classified as low-resource.

ID	Evidence
Q159	"More specifically, we want to answer the following two questions: (1) How well do models generalize to non-Wikimedia domains? (2) Does fine-tuning on high quality in-domain parallel text lead to good performance?" (p.24)
Q419	"In terms of resource level, there are 40 high-resource and 70 low-resource directions (see Table 1 and Table 51)." (p.49)
Q420	"Out of 70 low-resource directions, 22 are very low-resource, i.e., have less than 100K training examples." (p.49)

Checklist

Checklist item definitions:

ID	Assessment Question
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IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Required deployment categories include: agricultural input subsidy notices, land-use/land-tenure policy updates, microfinance credit-scheme terms, and bureaucratic/legal register documents with numbered clauses and conditional eligibility logic. None of these are represented in Flores-200 or NLLB-MD task taxonomy [Q95, Q166-Q169]; web search confirms no Ethiopian agricultural or administrative MT benchmark exists [WEB-3, WEB-4, WEB-10].

IO-2: Yes — the taxonomy omits all regionally relevant administrative/financial/agricultural categories. NLLB-MD covers only news, scripted talks, chat, and WHO health [Q166-Q169], and only for six languages not including Amharic [Q161]. The Flores-200 evaluation set is sourced from Wikinews, Wikijunior, and Wikivoyage [Q95].

IO-3: The benchmark includes ancillary subtasks (LID [Q179], bitext mining [Q286], toxicity [Q699]) that are upstream model-development tasks rather than irrelevant for the deployment per se, but no category is irrelevant in a way that consumes evaluation weight. Domain coverage in NLLB-MD (chat, scripted talks) is not aligned with the deployment but does not displace deployment-relevant evaluation since Amharic is excluded from NLLB-MD entirely [Q161].

IO-4: Category gaps that harm content validity: (a) no agricultural/subsidy genre, (b) no land-tenure/legal proclamation genre, (c) no microfinance/credit-agreement genre, (d) no document-level coherence or terminology-consistency task. These omissions are confirmed structural — not addressable by tuning thresholds [WEB-10 confirms domain-benchmark mismatch is real].

ID	Evidence
Q95	“As a significant extension of Flores-101, Flores-200 consists of 3001 sentences sampled from English-language Wikimedia projects for 204 total languages. Approximately one third of sentences are collected from each of these sources: Wikinews, Wikijunior, and Wikivoyage. The content is professionally translated into 200+ languages to create Flores-200. As we translate the same set of content into all languages, Flores-200 is a many-to-many multilingual benchmark.” (p.19)
Q159	“More specifically, we want to answer the following two questions: (1) How well do models generalize to non-Wikimedia domains? (2) Does fine-tuning on high quality in-domain parallel text lead to good performance?” (p.24)
Q161	“NLLB-MD covers the following six languages: Central Aymara (ayr_Latn), Bhojpuri (bho_Deva), Dyula (dyu_Latn), Friulian (fur_Latn), Russian (rus_Cyrl) and Wolof (wol_Latn).” (p.24)
Q166	“We translate the English side of the WMT21 English-German development set, containing a sample of newspapers from 2020 (Akhbardeh et al., 2021).” (p.25)
Q167	“We translate text extracted from a series of scripted English-language talks covering a variety of topics.” (p.25)
Q168	“We extract 3,000 utterances from the multi-session chat dataset of Xu et al. (2022), which contains on average 23 words per turn.” (p.25)
Q169	“We translated one World Health Organisation report (Donaldson and Rutter, 2017) and combined it with sentences translated from the English portion of the TAUS Corona Crisis Report.” (p.25)
Q179	“Language identification (LID) is the task of predicting the primary language for a span of texts.” (p.26)
Q286	“The underlying idea of our bitext mining approach is to first learn a multilingual sentence embedding space and to use a similarity measure in that space to decide whether two sentences are parallel or not.” (p.37)
Q699	“Our goal in this section is to provide an analysis of toxicity in multilingual MT models.” (p.79)
WEB-3	No Ethiopian agricultural or administrative-domain Amharic benchmark exists (https://arxiv.org/abs/2403.13737)
WEB-4	EthioMT parallel corpora (15 Ethiopian languages) drawn from religious and general domains, not administrative/financial (https://arxiv.org/html/2403.19365v1)
WEB-10	2025 study: mock translations merely reproducing named entities achieved non-trivial BLEU and ChrF++ scores, confirming metrics can reward superficial overlap rather than translational adequacy (https://arxiv.org/html/2508.20511v1)

Information Gaps

- Whether any private-sector or government-internal Amharic agricultural/financial MT benchmark exists; web search found none.

Requires Expert Verification

- Whether the document-genre distinctions matter as much in practice as the elicitation suggests, given that source documents are primarily already in Amharic (per regional context, the deployment is converting documents into Amharic).

Remediation

Gap 1: No agricultural, land-tenure, microfinance, or bureaucratic-legal genres in the task taxonomy; NLLB-MD covers only news/talks/chat/health and excludes Amharic [Q161, Q166-Q169].

Recommendation: Construct a deployment-specific evaluation set covering the three target document genres (subsidy notices, land-use updates, credit-scheme terms) with at least 200-500 sentences per genre, drawn from real federal/regional bureau and ACSI/ADCSI documents (with appropriate redaction).

ID	Evidence
Q161	"NLLB-MD covers the following six languages: Central Aymara (ayr_Latn), Bhojpuri (bho_Deva), Dyula (dyu_Latn), Friulian (fur_Latn), Russian (rus_Cyrl) and Wolof (wol_Latn)." (p.24)
Q166	"We translate the English side of the WMT21 English-German development set, containing a sample of newspapers from 2020 (Akhbardeh et al., 2021)." (p.25)
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Q168	"We extract 3,000 utterances from the multi-session chat dataset of Xu et al. (2022), which contains on average 23 words per turn." (p.25)
Q169	"We translated one World Health Organisation report (Donaldson and Rutter, 2017) and combined it with sentences translated from the English portion of the TAUS Corona Crisis Report." (p.25)

Input Content (IC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: NLLB | Context: Amhara Region Agricultural and Microfinance Translation Cohort

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Flores-200 and NLLB-Seed inputs are exclusively sampled from English-language Wikimedia projects [Q95, Q146], and the paper explicitly acknowledges 'the content reflects what Wikipedia editors find is relevant for English Wikipedia, and likely does not cover diverse content from different cultures' [Q151]. Low-resource bitext for African languages is 'often opportunistically drawn from known collections of multilingual content, such as the Christian Bible or publications of multinational organizations' [Q284], 'often limited in quantity and domain' [Q285]. The deployment requires Ethiopian-specific land-tenure proclamation vocabulary, ACSI/ADCSI financial terminology, and regional cooperative administrative phrasing — none of which is documented as present in NLLB training or evaluation data. The Hugging Face NLLB model card confirms 'Our model has been tested on the Wikimedia domain with limited investigation on other domains' [WEB-11]. Web search found no public ACSI/ADCSI Amharic terminology resources [WEB-15, WEB-16], confirming the absence of any reference content for the deployment's core terminology. Given HIGH priority and binding-document consequences, the content gap is severe.

ID	Evidence
Q95	"As a significant extension of Flores-101, Flores-200 consists of 3001 sentences sampled from English-language Wikimedia projects for 204 total languages. Approximately one third of sentences are collected from each of these sources: Wikinews, Wikijunior, and Wikivoyage. The content is professionally translated into 200+ languages to create Flores-200. As we translate the same set of content into all languages, Flores-200 is a many-to-many multilingual benchmark." (p.19)
Q146	"Data for NLLB-Seed was sampled from Wikimedia's List of articles every Wikipedia should have, a collection of 10,000 Wikidata IDs corresponding to notable topics in different fields of knowledge and human activity. These are split into 11 categories such as People, History, Philosophy and Religion, Geography." (p.23)
Q151	"We note that like Flores-200, NLLB-Seed's source data is English-centric and sampled from English Wikipedia. This has an important effect: the content reflects what Wikipedia editors find is relevant for English Wikipedia, and likely does not cover diverse content from different cultures." (p.24)
Q284	"Existing parallel corpora for low-resource languages are often opportunistically drawn from known collections of multilingual content, such as the Christian Bible or publications of multinational organizations." (p.37)
Q285	"These are often limited in quantity and domain." (p.37)
WEB-11	NLLB Hugging Face model card: 'Our model has been tested on the Wikimedia domain with limited investigation on other domains' (https://huggingface.co/facebook/nllb-200-3.3B)
WEB-15	ACSI official website — no Amharic financial terminology guide publicly available (http://www.acsi.org.et/)
WEB-16	FinDev Gateway ACSI profile — no terminology resource (https://www.findevgateway.org/case-study/2004/01/amhara-credit-and-saving-institution-acsi-institutional-profile-current-status)

Strengths

- Deliberate Amharic support: dedicated 8,000-token Ge'ez-script SentencePiece vocabulary [Q382] and combined supervised cosine-loss + unsupervised MLM training [Q373, Q374], demonstrating attention to Amharic representation at the modeling level.
- Acknowledged limitation: the paper transparently flags the English-Wikipedia-centric content bias [Q151], allowing users to factor it into deployment decisions.
- Cross-referencing with Ethnologue and Glottolog [Q80, Q81] provides linguistic metadata that confirms language identity, even if content remains Wikimedia-derived.

ID	Evidence
Q80	"The language names have been cross-referenced with major linguistic information platforms such as Ethnologue (Lewis, 2009) and Glottolog (Hammarström et al., 2022)." (p.17)
Q81	"We provide Language family information for each language based on the Glottolog database (Hammarström et al., 2022)." (p.17)
Q151	"We note that like Flores-200, NLLB-Seed's source data is English-centric and sampled from English Wikipedia. This has an important effect: the content reflects what Wikipedia editors find is relevant for English Wikipedia, and likely does not cover diverse content from different cultures." (p.24)
Q373	"We also deployed a new training procedure which combines supervised training, i.e. minimizing the cosine loss between the teacher and student embedding, and unsupervised masked LM training (see left part of Figure 12)." (p.44)
Q374	"This enabled us to benefit from monolingual data during encoder training." (p.44)
Q382	"We then trained an encoder for Amharic and Tigrinya only, paired with English as in all our experiments, and a specific 8k SentencePiece vocabulary to better support the Ge'ez script." (p.45)

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Yes — the deployment requires region-specific cultural, legal-administrative, and financial knowledge: Ethiopian federal land proclamation vocabulary, ACSI/ADCSI loan terminology, cooperative-tier benefit language. The benchmark's content is sourced from English Wikinews/Wikijunior/Wikivoyage [Q95] and contains no documented Ethiopian administrative content.

IC-2: The benchmark's culturally sensitive content (toxicity wordlists, etc.) is not regionally calibrated for Ethiopian context [Q715, Q716]. The general-domain Wikimedia content does not align with the deployment's bureaucratic/financial register.

IC-3: Inputs derive from English-Wikipedia-relevance judgments [Q151], which encode topic priorities (Western-perspective biographies, geography, philosophy [Q146]) that do not transfer to Ethiopian agricultural/microfinance content.

IC-4: INSUFFICIENT DOCUMENTATION — the registry contains no record of Ethiopian regional annotators reviewing Amharic input content for cultural specificity. Web search [WEB-13, WEB-14] documents post-hoc corrections for several African Flores-200 splits (Hausa, Sepedi, Xitsonga, isiZulu) but none for Amharic.

IC-5: Content issues harming validity: (a) zero coverage of Ethiopian agricultural/financial vocabulary [WEB-15, WEB-16]; (b) zero coverage of federal land proclamation terminology; (c) Wikimedia content bias acknowledged by authors [Q151]; (d) low-resource African bitext relies on Bible and multinational publications [Q284] — domains divergent from the deployment.

ID	Evidence
Q95	"As a significant extension of Flores-101, Flores-200 consists of 3001 sentences sampled from English-language Wikimedia projects for 204 total languages. Approximately one third of sentences are collected from each of these sources: Wikinews, Wikijunior, and Wikivoyage. The content is professionally translated into 200+ languages to create Flores-200. As we translate the same set of content into all languages, Flores-200 is a many-to-many multilingual benchmark." (p.19)
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Q151	"We note that like Flores-200, NLLB-Seed's source data is English-centric and sampled from English Wikipedia. This has an important effect: the content reflects what Wikipedia editors find is relevant for English Wikipedia, and likely does not cover diverse content from different cultures." (p.24)

Q284	“Existing parallel corpora for low-resource languages are often opportunistically drawn from known collections of multilingual content, such as the Christian Bible or publications of multinational organizations.” (p.37)
Q285	“These are often limited in quantity and domain.” (p.37)
Q715	“Toxicity is culturally sensitive, which constitutes a challenge when starting from one source language (in this case, English) and attempting to find equivalents in such a largely multilingual setting.” (p.80)
Q716	“Hate speech terms such as racial or ethnic slurs, for example, may well be the most challenging of all.” (p.80)
WEB-3	No Ethiopian agricultural or administrative-domain Amharic benchmark exists (https://arxiv.org/abs/2403.13737)
WEB-13	Nature NLLB paper confirms general workflow but no Amharic-specific domain expertise documentation (https://www.nature.com/articles/s41586-024-07335-x)
WEB-14	Post-publication FLORES-200 corrections exist for Hausa, Sepedi, Xitsonga, isiZulu — but not Amharic, leaving Amharic reference unaudited (https://www.emergentmind.com/topics/flores-200-benchmark-dataset)
WEB-15	ACSI official website — no Amharic financial terminology guide publicly available (http://www.acsi.org.et/)
WEB-16	FinDev Gateway ACSI profile — no terminology resource (https://www.findevgateway.org/case-study/2004/01/amhara-credit-and-saving-institution-acsi-institutional-profile-current-status)

Information Gaps

- No corpus-level analysis of NLLB training data is publicly available to confirm or deny presence of Ethiopian government administrative vocabulary [WEB-11].
- No Amharic-specific Flores-200 audit of error rates exists [WEB-14].

Requires Expert Verification

- Whether ACSI/ADCSI internal terminology actually diverges from federally standardized vocabulary, or whether federal standardization is sufficient — registry flags this as 'NEEDS VERIFICATION'.
- Whether domain-specific terminology gaps are recoverable through fine-tuning or require rebuilt evaluation data.

Remediation

Gap 1: Wikimedia-derived English-centric content [Q95, Q151] contains no Ethiopian agricultural/financial/administrative vocabulary; ACSI/ADCSI publish no public terminology resources [WEB-15, WEB-16].

Recommendation: Build a controlled Ethiopian administrative terminology lexicon covering federal land proclamation terms, ACSI/ADCSI loan vocabulary, and cooperative-tier benefits — sourced from federal/regional bureaus and validated by domain translators — and use it both as fine-tuning augmentation and as a glossary check during evaluation.

Gap 2: No public ACSI/ADCSI terminology guide [WEB-15, WEB-16] and no domain reference data exists for validation [WEB-3, WEB-4].

Recommendation: Engage ACSI/ADCSI directly through a memorandum of understanding to access internal Amharic terminology references and historical borrower-facing communications; treat these as the institutional gold standard for terminology rendering.

ID	Evidence
Q95	“As a significant extension of Flores-101, Flores-200 consists of 3001 sentences sampled from English-language Wikimedia projects for 204 total languages. Approximately one third of sentences are collected from each of these sources: Wikinews, Wikijunior, and Wikivoyage. The content is professionally translated into 200+ languages to create Flores-200. As we translate the same set of content into all languages, Flores-200 is a many-to-many multilingual benchmark.” (p.19)
Q151	“We note that like Flores-200, NLLB-Seed’s source data is English-centric and sampled from English Wikipedia. This has an important effect: the content reflects what Wikipedia editors find is relevant for English Wikipedia, and likely does not cover diverse content from different cultures.” (p.24)
WEB-3	No Ethiopian agricultural or administrative-domain Amharic benchmark exists (https://arxiv.org/abs/2403.13737)
WEB-4	EthioMT parallel corpora (15 Ethiopian languages) drawn from religious and general domains, not administrative/financial (https://arxiv.org/html/2403.19365v1)

WEB-15	ACSI official website — no Amharic financial terminology guide publicly available (http://www.acsi.org.et/)
WEB-16	FinDev Gateway ACSI profile — no terminology resource (https://www.findevgateway.org/case-study/2004/01/amhara-credit-and-saving-institution-acsi-institutional-profile-current-status)

Input Form (IF)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: NLLB | Context: Amhara Region Agricultural and Microfinance Translation Cohort

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The benchmark's input form is well-aligned with the deployment at the modality and script levels. All inputs are text-only encoded via BCP 47 with ISO 639-3 + ISO 15924 script subtags [\[Q79\]](#), the Ethiopic (Ge'ez) script is explicitly supported with a dedicated 8k SentencePiece vocabulary [\[Q382\]](#), and the deployer confirmed text-only prose input is acceptable [elicitation A4]. Average sentence length (~21 words) [\[Q133\]](#) reflects general-domain prose. The deployment is HIGH on text-only operation, and the benchmark matches. Residual concerns are: (a) sentence-level evaluation architecture does not assess document-level structures (numbered clauses, conditional logic), and (b) Flores-200 sentences are general-domain prose, whereas administrative source documents have higher punctuation density and clause structure — though deployer accepted prose output [elicitation A4]. IF was assigned LOWER priority by elicitation, consistent with this score.

ID	Evidence
Q79	"We represent each language with a BCP 47 tag sequence using a three-letter ISO 639-3 code as the base subtag, which we complement with ISO 15924 script subtags, as we collected resources for several languages in more than one script." (p.17)
Q133	"On average, sentences are approximately 21 words long." (p.22)
Q382	"We then trained an encoder for Amharic and Tigrinya only, paired with English as in all our experiments, and a specific 8k SentencePiece vocabulary to better support the Ge'ez script." (p.45)

Strengths

- Dedicated 8,000-token Ge'ez SentencePiece vocabulary trained jointly for Amharic and Tigrinya [\[Q382\]](#) addresses a key low-resource representation issue.
- BCP 47 / ISO 639-3 / ISO 15924 standardization [\[Q79\]](#) gives unambiguous language and script identification, important since Amharic uses Ethiopic exclusively.
- Combined supervised cosine-loss + unsupervised MLM training procedure [\[Q373, Q374\]](#) leverages monolingual Amharic data, which is more plentiful than bitext.

ID	Evidence
Q79	"We represent each language with a BCP 47 tag sequence using a three-letter ISO 639-3 code as the base subtag, which we complement with ISO 15924 script subtags, as we collected resources for several languages in more than one script." (p.17)
Q373	"We also deployed a new training procedure which combines supervised training, i.e. minimizing the cosine loss between the teacher and student embedding, and unsupervised masked LM training (see left part of Figure 12)." (p.44)
Q374	"This enabled us to benefit from monolingual data during encoder training." (p.44)
Q382	"We then trained an encoder for Amharic and Tigrinya only, paired with English as in all our experiments, and a specific 8k SentencePiece vocabulary to better support the Ge'ez script." (p.45)

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target

IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Signal distributions broadly match: text-only, Ethiopic script, sentence-level prose [Q79, Q133]. Source documents in deployment are also text in Ethiopic script. Sentence length distribution may differ (administrative documents have longer, clause-heavy sentences) but benchmark sentences average ~21 words [Q133] which is in a reasonable range.

IF-2: Regional infrastructure can support text input; the deployment is text-only. However, web search [WEB-7, WEB-8] documents recurring mobile internet shutdowns in Amhara (August 2023–July 2024 and April–May 2023) which could disrupt connected pipelines, but this is an operational risk rather than a benchmark-form mismatch.

IF-3: Domain-specific form differences: administrative source documents contain numbered clauses, tables, and cross-references absent from Flores-200 sentences [Q95, Q133]. The benchmark does not evaluate document-level coherence. The deployer accepted prose output [elicitation A4], reducing the severity of this gap.

IF-4: Form mismatch is minor at the script/modality level. The principal residual gap is the absence of document-level evaluation, which is more accurately a scoring/output issue than an input-form issue.

ID	Evidence
Q79	"We represent each language with a BCP 47 tag sequence using a three-letter ISO 639-3 code as the base subtag, which we complement with ISO 15924 script subtags, as we collected resources for several languages in more than one script." (p.17)
Q95	"As a significant extension of Flores-101, Flores-200 consists of 3001 sentences sampled from English-language Wikimedia projects for 204 total languages. Approximately one third of sentences are collected from each of these sources: Wikinews, Wikijunior, and Wikivoyage. The content is professionally translated into 200+ languages to create Flores-200. As we translate the same set of content into all languages, Flores-200 is a many-to-many multilingual benchmark." (p.19)
Q133	"On average, sentences are approximately 21 words long." (p.22)
WEB-7	Freedom House 2024: documented Amhara mobile internet shutdowns Aug 2023–July 2024 (operational risk, not benchmark form mismatch) (https://freedomhouse.org/country/ethiopia/freedom-net/2024)
WEB-8	Freedom House 2023: prior April–May 2023 connectivity blackouts in Amhara (https://freedomhouse.org/country/ethiopia/freedom-net/2023)

Information Gaps

- Sentence-length distribution comparison between Flores-200 and actual administrative Amharic documents is not documented.

Output Ontology (OO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: NLLB | Context: Amhara Region Agricultural and Microfinance Translation Cohort

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The benchmark's output scoring ontology is a uniform scalar quality score (chrF++, spBLEU, or calibrated XSTS) applied to all directions and sentence types [Q455, Q648, Q661]. There is no domain-specific rubric for correctly rendering numerical eligibility conditions, conditional legal clauses, or financial terminology — the most consequential error types for the deployment. The XSTS scale measures 'meaning preservation' on a 5-point scale [Q654] but treats sentences uniformly. For toxicity, the taxonomy distinguishes source-present from added toxicity [Q705, Q706] with categories covering vulgar/profane/pornographic/hate-speech/bullying [Q712-Q714]; the paper acknowledges this is 'culturally sensitive' [Q715] with no Ethiopian regional calibration. A chrF++ improvement of +0.5 corresponds to detectable human improvement [Q624, Q631] but cannot distinguish silently altered subsidy thresholds from generally good prose. Independent 2025 evidence [WEB-10] shows mock translations reproducing named entities scored non-trivially on BLEU/ChrF++, confirming these metrics reward superficial overlap. Given OO is HIGH priority for binding documents, this is a structural-validity violation.

ID	Evidence
Q455	"We use the chrF++ metric to compare the model performance (see Section 7.1)." (p.53)
Q624	"We investigate the relationship between chrF++ score and human evaluation score to understand what quantity of automatic metric improvement in a model ablation overall, we find that chrF++ improvements of +0.5 chrF++ usually correlate to statistically significant human evaluation improvements, with a score of +1 chrF++ almost always being detectable by human evaluators." (p.70)
Q631	"In conclusion, we believe that based on our human evaluation studies of model ablations, that an improvement of +0.5 chrF++ is often detectable by human evaluators." (p.71)
Q648	"In this work, we primarily evaluate using chrF++ using the settings from sacrebleu. However, when comparing to other published works, we utilize BLEU and spBLEU where appropriate." (p.73)
Q654	"XSTS rates each source sentence and its machine translation on a five-point scale, where 1 is the lowest score and 5 is the highest score." (p.74)
Q661	"To obtain an aggregate human quality metric for each language direction in an evaluation study, we take the majority XSTS score for each sentence and average these majority scores over all evaluated sentences." (p.74)
Q705	"In the context of translation, toxicity may originally be present in the source text or it can be generated de-novo in the target text (added toxicity)." (p.79)
Q706	"This added toxicity can come from a mistranslation (e.g., wrong lexical choice) or as a hallucination of a new target word from zero — both produce inaccurate translations." (p.79)
Q712	"We include items that are commonly referred to as vulgar or profane language." (p.80)
Q713	"In addition to these, we include items more specifically associated with depictions of pornographic content or sexual acts, some frequently used hate speech expressions, and some bullying expressions (including language that can cause trauma or be used with the purpose of silencing someone)." (p.80)
Q714	"We also include items, vulgar or not, referring to body parts that are commonly associated with sexual practices." (p.80)
Q715	"Toxicity is culturally sensitive, which constitutes a challenge when starting from one source language (in this case, English) and attempting to find equivalents in such a largely multilingual setting." (p.80)
WEB-10	2025 study: mock translations merely reproducing named entities achieved non-trivial BLEU and ChrF++ scores, confirming metrics can reward superficial overlap rather than translational adequacy (https://arxiv.org/html/2508.20511v1)

Strengths

- XSTS protocol with calibration sets [Q650, Q664] addresses cross-language comparability of human evaluations, a non-trivial methodological contribution.
- Toxicity ontology distinguishing source-present vs. added toxicity [Q705, Q706] is conceptually appropriate for safety and partially relevant to the deployment's high-stakes context.
- The paper rejects unreliable model-based and roundtrip metrics [Q638, Q639] and explains its choice of chrF++ as the primary metric, providing transparency.

ID	Evidence
Q638	"While model-based metrics have shown better correlation with human judgment in recent metrics shared tasks (Freitag et al., 2021), they require training and are not easily extended to a large set of low-resource languages." (p.72)
Q639	"Another approach is to use highly approximate metrics based on roundtrip translations such as RttLangIDChrF (Bapna et al., 2022), although roundtrip translation may not correlate well with translation quality (Koehn, 2005)." (p.72)
Q650	"We use two advances — the XSTS evaluation protocol and the use of calibration sets — to enable meaningful human evaluation scores that are comparable across language pairs." (p.73)
Q664	"The calibration sets for all evaluation studies are drawn from a set of source, target sentence pairs consisting of K = 1000 backtranslated Flores-200 sentences of varying quality." (p.75)
Q705	"In the context of translation, toxicity may originally be present in the source text or it can be generated de-novo in the target text (added toxicity)." (p.79)
Q706	"This added toxicity can come from a mistranslation (e.g., wrong lexical choice) or as a hallucination of a new target word from zero — both produce inaccurate translations." (p.79)

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: The output label categories (chrF++, spBLEU, XSTS 1-5) [Q455, Q654, Q648] are general translation-quality scores. They are not designed to flag deployment-critical errors such as misrendered eligibility thresholds, altered penalty rates, or inconsistent terminology across clauses.

OO-2: Missing categories specific to the deployment: (a) numerical-fidelity scoring (correct rendering of percentages, dates, monetary amounts in legal clauses); (b) terminology-consistency scoring across a document; (c) conditional-logic preservation scoring; (d) administrative-register adherence scoring. None of these exist in NLLB's output ontology [Q455, Q648, Q654].

OO-3: The toxicity taxonomy is acknowledged as culturally sensitive [Q715, Q716] with English-anchored start [Q717], and no Ethiopian-specific calibration is documented. This may encode source-culture defaults but is a secondary concern relative to the missing domain-correctness rubric.

OO-4: Stakeholder-driven taxonomy redesign would be required: federal bureau translators (deployer's stated authority [elicitation A3]) would need to define a domain rubric distinguishing acceptable from catastrophic translation error types for subsidy/credit/land documents.

OO-5: Taxonomy issues harming validity: (a) structural validity — output ontology does not represent the construct of 'fitness for binding administrative documents'; (b) content validity — categories for high-consequence error types are absent; (c) external validity — chrF++ improvements do not predict deployment fitness [WEB-10].

ID	Evidence
Q455	"We use the chrF++ metric to compare the model performance (see Section 7.1)." (p.53)
Q648	"In this work, we primarily evaluate using chrF++ using the settings from sacrebleu. However, when comparing to other published works, we utilize BLEU and spBLEU where appropriate." (p.73)
Q654	"XSTS rates each source sentence and its machine translation on a five-point scale, where 1 is the lowest score and 5 is the highest score." (p.74)
Q661	"To obtain an aggregate human quality metric for each language direction in an evaluation study, we take the majority XSTS score for each sentence and average these majority scores over all evaluated sentences." (p.74)
Q715	"Toxicity is culturally sensitive, which constitutes a challenge when starting from one source language (in this case, English) and attempting to find equivalents in such a largely multilingual setting." (p.80)
Q716	"Hate speech terms such as racial or ethnic slurs, for example, may well be the most challenging of all." (p.80)
Q717	"We begin based on the professional translation of an initial list assembled in English, and allow additions to adapt to cultural specificities." (p.80)
WEB-10	2025 study: mock translations merely reproducing named entities achieved non-trivial BLEU and ChrF++ scores, confirming metrics can reward superficial overlap rather than translational adequacy (https://arxiv.org/html/2508.20511v1)

Information Gaps

- No documented case study of NLLB mistranslating a numerical legal-financial clause in Amharic specifically [WEB-10]; the gap is inferential from metric design rather than empirically demonstrated for this language.

Requires Expert Verification

- Federal bureau translators' specification of which error categories should be tracked for subsidy/credit document QA — required to design any deployment-specific rubric.

Remediation

Gap 1: Scoring is uniform scalar chrF++/spBLEU/XSTS [Q455, Q648, Q654] with no rubric for numerical-fidelity, conditional-logic preservation, or terminology consistency — the highest-consequence error types for binding documents.

Recommendation: Develop a deployment-specific error rubric (in collaboration with federal bureau translators) that flags catastrophic categories: (a) altered numerical thresholds or dates, (b) reversed/dropped conditional clauses, (c) inconsistent rendering of defined terms across the document, (d) loss of named entities. Score translations against this rubric in addition to chrF++.

ID	Evidence
Q455	"We use the chrF++ metric to compare the model performance (see Section 7.1)." (p.53)
Q648	"In this work, we primarily evaluate using chrF++ using the settings from sacrebleu. However, when comparing to other published works, we utilize BLEU and spBLEU where appropriate." (p.73)
Q654	"XSTS rates each source sentence and its machine translation on a five-point scale, where 1 is the lowest score and 5 is the highest score." (p.74)

Output Content (OC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: NLLB | Context: Amhara Region Agricultural and Microfinance Translation Cohort

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Flores-200 reference translations were produced through a multi-step workflow [Q93] with translators requiring 2-5 years of experience [Q97], 18-month re-testing [Q98], and reviewer native-speaker status with typically 5+ years of experience [Q99, Q100]. For low-resource languages, qualification thresholds were relaxed to accept broader linguistics degrees or extensive commercial experience [Q103]. No Amharic-specific annotator provenance, domain expertise, or inter-annotator agreement figures are documented in the registry. The community interview study informing research priorities included only 12 African-language speakers [Q31], a majority being immigrants in US/Europe with ~1/3 tech workers [Q32], whose perspectives 'may not consummately capture the sentiments of their communities back home' [Q39]. The deployer's required validation authority — federal bureau translators with Ethiopian agricultural/financial domain expertise [elicitation A3] — is not represented in the Flores-200 annotator workforce. Web search confirms post-publication corrections exist for Hausa/Sepedi/Xitsonga/isiZulu Flores-200 splits [WEB-14] but not Amharic, leaving Amharic reference quality unaudited. Common errors documented include 'mistranslation and unnatural translation errors' from non-English language influences and 'lower levels of standardization' [Q136-Q138]. Given OC is HIGH priority, this is a substantial concern.

ID	Evidence
Q31	"Covering a total of 36 languages, the distribution is as follows: 5 languages are spoken predominantly in North America, 8 in South America, 4 in Europe, 12 in Africa, and 7 in Asia." (p.8)
Q32	"Although our sample has breadth in terms of race, education, and location, the majority of our participants are immigrants living in the U.S. and Europe, and about a third of them (n = 17) identify as tech workers." (p.8)
Q39	"For one, as immigrants, their perspectives may not consummately capture the sentiments of their communities back home." (p.8)
Q93	"We created a complex, multi-step process to ensure quality. First, professional translators and reviewers aligned on language standards. Next, translators translated the full set of Flores-200 sentences, followed by automated checks. Subsequently, the group of independent reviewers reviewed the quality, and based on their assessment, we sent some translations out for post-editing. If the quality assessment indicated that the quality is above 90 percent, the language is considered ready for inclusion in Flores-200." (p.19)
Q97	"Translators are required to have at least two to three years of translation experience in the relevant language pair if they have an academic degree in translation or linguistics and three to five years of translation experience if they do not have any relevant academic qualification." (p.20)
Q98	"Translators also undergo a translation test every 18 months to assess their translation quality." (p.20)
Q99	"Flores-200 reviewers are also required to be native speakers of the target language." (p.20)
Q100	"Reviewers typically have a translation degree, at least five years of experience working as a translator, translation review experience, and where possible are accredited by a relevant translation board." (p.20)
Q103	"In these cases, we modified the qualification process to accept applications from reviewers with more general language degrees such as Linguistics or African Language Studies, or no degree provided they have had extensive commercial translation experience (e.g. >10 years)." (p.20)
Q136	"Mistranslation and unnatural translation errors were still the most common errors found while assessing the quality of the human translations." (p.22)
Q137	"These were mainly due to influences from other non-English languages that may be prominently used in the target communities, leading to excessive borrowings of vocabulary and grammar, literal translations due to infrequent usage of the target language in a formal, informative content style and the lower levels of standardization." (p.22)
Q138	"There has also been an increasing trend in spelling inconsistencies in the human translations due to lower levels of standardization leading to inconsistent or even subjective or preferential approaches." (p.22)

WEB-14	Post-publication FLORES-200 corrections exist for Hausa, Sepedi, Xitsonga, isiZulu — but not Amharic, leaving Amharic reference unaudited (https://www.emergentmind.com/topics/flores-200-benchmark-dataset)
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Strengths

- Documented multi-step QA workflow with 90% quality threshold [Q93, Q115, Q135] and post-editing for sub-threshold languages provides a baseline quality floor.
- Native-speaker reviewer requirement [Q99] preserves a basic legitimacy criterion for translation correctness.
- Transparency about diaspora bias in the community study [Q38, Q39, Q40] allows users to factor in selection bias when interpreting community-grounded claims.

ID	Evidence
Q38	"We acknowledge that sampling low-resource language speakers from diasporic contexts comes with its limitations." (p.8)
Q39	"For one, as immigrants, their perspectives may not consummately capture the sentiments of their communities back home." (p.8)
Q40	"Over-sampling tech workers may introduce another form of selection bias." (p.8)
Q93	"We created a complex, multi-step process to ensure quality. First, professional translators and reviewers aligned on language standards. Next, translators translated the full set of Flores-200 sentences, followed by automated checks. Subsequently, the group of independent reviewers reviewed the quality, and based on their assessment, we sent some translations out for post-editing. If the quality assessment indicated that the quality is above 90 percent, the language is considered ready for inclusion in Flores-200." (p.19)
Q99	"Flores-200 reviewers are also required to be native speakers of the target language." (p.20)
Q115	"When full translation is completed, the QA LSP team performs a final QA review and assesses a 20% sample data. Optional arbitration, rework and QA spot checks may follow if the final quality score of the translation dataset is below 90%." (p.21)
Q135	"The minimum acceptable standard is 90 percent." (p.22)

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: Ground-truth labels are general-translation-quality judgments by professional translators with 2+ years' experience [Q97]. They do not reflect Ethiopian federal bureau translator perspectives on agricultural/financial document correctness — the deployer's stated authority [elicitation A3].

OC-2: High potential for disagreement: the deployer requires domain-trained federal bureau translators as primary authority [elicitation A3], whereas Flores-200 annotators are general translators without documented Ethiopian government-domain expertise [Q97-Q103]. Domain-specific terminology choices (e.g., loan-penalty rendering) likely differ.

OC-3: Annotator demographics for Amharic specifically are not documented in the verbatim quote registry. General requirements [Q97-Q103] are documented but no Amharic-specific breakdown exists. Web search [WEB-13, WEB-14] confirms Nature 2024 paper documents the general workflow but no Amharic-specific annotator details.

OC-4: Re-annotation by representative regional annotator pool would be required for deployment validation. No such pool is documented; ACSI/ADCSI publish no terminology guides [WEB-15, WEB-16] that could serve as gold standard.

OC-5: Aggregation uses majority XSTS score per sentence then averaging [Q661]; this could erase minority perspectives, but for translation quality the more relevant concern is that minority-but-correct domain-specific renderings could be averaged out.

OC-6: Label issues harming validity: (a) convergent validity — annotator pool does not converge on Ethiopian government-domain perspective [elicitation A3]; (b) external validity — Flores-200 quality judgments unlikely to predict acceptance by federal bureau translators or cooperative end-users with established administrative vocabulary.

ID	Evidence
Q97	"Translators are required to have at least two to three years of translation experience in the relevant language pair if they have an academic degree in translation or linguistics and three to five years of translation experience if they do not have any relevant academic qualification." (p.20)
Q98	"Translators also undergo a translation test every 18 months to assess their translation quality." (p.20)
Q99	"Flores-200 reviewers are also required to be native speakers of the target language." (p.20)
Q100	"Reviewers typically have a translation degree, at least five years of experience working as a translator, translation review experience, and where possible are accredited by a relevant translation board." (p.20)
Q101	"We note that these are stringent standards, and extensions of Flores-200 to even more low-resource languages in the future may be difficult." (p.20)
Q102	"Already for many languages, finding a reviewer that meets the criteria above is very challenging." (p.20)
Q103	"In these cases, we modified the qualification process to accept applications from reviewers with more general language degrees such as Linguistics or African Language Studies, or no degree provided they have had extensive commercial translation experience (e.g. >10 years)." (p.20)
Q136	"Mistranslation and unnatural translation errors were still the most common errors found while assessing the quality of the human translations." (p.22)
Q137	"These were mainly due to influences from other non-English languages that may be prominently used in the target communities, leading to excessive borrowings of vocabulary and grammar, literal translations due to infrequent usage of the target language in a formal, informative content style and the lower levels of standardization." (p.22)
Q661	"To obtain an aggregate human quality metric for each language direction in an evaluation study, we take the majority XSTS score for each sentence and average these majority scores over all evaluated sentences." (p.74)
WEB-13	Nature NLLB paper confirms general workflow but no Amharic-specific domain expertise documentation (https://www.nature.com/articles/s41586-024-07335-x)
WEB-14	Post-publication FLORES-200 corrections exist for Hausa, Sepedi, Xitsonga, isiZulu — but not Amharic, leaving Amharic reference unaudited (https://www.emergentmind.com/topics/flores-200-benchmark-dataset)
WEB-15	ACSI official website — no Amharic financial terminology guide publicly available (http://www.acsi.org.et/)
WEB-16	FinDev Gateway ACSI profile — no terminology resource (https://www.findevgateway.org/case-study/2004/01/amhara-credit-and-saving-institution-acsi-institutional-profile-current-status)

Information Gaps

- No public record of who specifically annotated the Amharic Flores-200 split, their credentials, or their domain backgrounds.
- No published Amharic-specific Flores-200 error audit comparable to those for Hausa/Sepedi/Xitsonga/isiZulu [WEB-14].

Requires Expert Verification

- Whether federal bureau translators would actually disagree with Flores-200 Amharic reference renderings on general-domain Wikimedia content (a separate question from domain-specific document fitness).

Remediation

Gap 1: Amharic Flores-200 annotator credentials and provenance not documented [no Q-evidence]; annotators are general professional translators [Q97-Q103], not Ethiopian government-domain specialists; no Amharic-specific post-publication audit exists [WEB-14].

Recommendation: Commission an Amharic Flores-200 audit analogous to the Abdulmumin et al. corrections for other African languages, and recruit a panel of federal bureau translators to re-annotate a domain-specific evaluation sample, treating their judgments as the convergent-validity gold standard.

ID	Evidence
Q97	“Translators are required to have at least two to three years of translation experience in the relevant language pair if they have an academic degree in translation or linguistics and three to five years of translation experience if they do not have any relevant academic qualification.” (p.20)
Q98	“Translators also undergo a translation test every 18 months to assess their translation quality.” (p.20)
Q99	“Flores-200 reviewers are also required to be native speakers of the target language.” (p.20)
Q100	“Reviewers typically have a translation degree, at least five years of experience working as a translator, translation review experience, and where possible are accredited by a relevant translation board.” (p.20)
Q101	“We note that these are stringent standards, and extensions of Flores-200 to even more low-resource languages in the future may be difficult.” (p.20)
Q102	“Already for many languages, finding a reviewer that meets the criteria above is very challenging.” (p.20)
Q103	“In these cases, we modified the qualification process to accept applications from reviewers with more general language degrees such as Linguistics or African Language Studies, or no degree provided they have had extensive commercial translation experience (e.g. >10 years).” (p.20)
WEB-14	Post-publication FLORES-200 corrections exist for Hausa, Sepedi, Xitsonga, isiZulu — but not Amharic, leaving Amharic reference unaudited (https://www.emergentmind.com/topics/flores-200-benchmark-dataset)

Output Form (OF)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: NLLB | Context: Amhara Region Agricultural and Microfinance Translation Cohort

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The benchmark evaluates text outputs only [Q5] using sentence-level chrF++ and spBLEU [Q455, Q645, Q648] plus 5-point XSTS [Q654]. The deployer confirmed text-only prose output is acceptable for this NLLB deployment [elicitation A4], and this matches the benchmark's output form. However, no document-level coherence, terminology-consistency, or structural-preservation metrics are reported, all evaluation is sentence-level on general-domain content [Q5, Q336], and aggregate sentence-level chrF++ improvements may not indicate fitness for binding administrative documents requiring consistent rendering across clauses. National Ethiopia adult literacy is ~51.8% [WEB-5] with rural Amhara likely lower, meaning translated documents are often relayed orally through cooperative leaders — a use case that further increases the importance of consistent terminology, which sentence-level scoring does not assess. OF was rated LOWER priority by elicitation; given form alignment at the modality level, a moderately high score is warranted, with the residual concern being absence of document-level scoring.

ID	Evidence
Q5	"we evaluated the performance of over 40,000 different translation directions using a human-translated benchmark, Flores-200, and combined human evaluation with a novel toxicity benchmark covering all languages in Flores-200 to assess translation safety." (p.1)
Q336	"We look for best performance on the Flores-200 development set, and report detokenized BLEU on Flores-200 devtest." (p.41)
Q455	"We use the chrF++ metric to compare the model performance (see Section 7.1)." (p.53)
Q645	"Goyal et al. (2022) propose spBLEU, a BLEU metric based on a standardized SentencePiece model (SPM) covering 101 languages, released with Flores101. In this work, we provide SPM-200 along with Flores-200 to enable measurement of spBLEU." (p.73)
Q648	"In this work, we primarily evaluate using chrF++ using the settings from sacrebleu. However, when comparing to other published works, we utilize BLEU and spBLEU where appropriate." (p.73)
Q654	"XSTS rates each source sentence and its machine translation on a five-point scale, where 1 is the lowest score and 5 is the highest score." (p.74)
WEB-5	Ethiopia adult literacy ~51.8% (UNESCO via World Bank, 2017) — establishes oral-relay relevance for output use (https://data.worldbank.org/indicator/SE.ADT.LITR.ZS?locations=ET)

Strengths

- Text-only prose output [Q5] aligns directly with the deployer's confirmed acceptable format [elicitation A4].
- chrF++ as primary metric [Q648] with documented +0.5 detectability threshold [Q624, Q631] provides interpretable signal.
- Use of standardized SPM-200 for spBLEU [Q645] supports cross-system comparability.

ID	Evidence
Q5	"we evaluated the performance of over 40,000 different translation directions using a human-translated benchmark, Flores-200, and combined human evaluation with a novel toxicity benchmark covering all languages in Flores-200 to assess translation safety." (p.1)
Q624	"We investigate the relationship between chrF++ score and human evaluation score to understand what quantity of automatic metric improvement in a model ablation overall, we find that chrF++ improvements of +0.5 chrF++ usually correlate to statistically significant human evaluation improvements, with a score of +1 chrF++ almost always being detectable by human evaluators." (p.70)

Q631	"In conclusion, we believe that based on our human evaluation studies of model ablations, that an improvement of +0.5 chrF++ is often detectable by human evaluators." (p.71)
Q645	"Goyal et al. (2022) propose spBLEU, a BLEU metric based on a standardized SentencePiece model (SPM) covering 101 languages, released with Flores101. In this work, we provide SPM-200 along with Flores-200 to enable measurement of spBLEU." (p.73)
Q648	"In this work, we primarily evaluate using chrF++ using the settings from sacrebleu. However, when comparing to other published works, we utilize BLEU and spBLEU where appropriate." (p.73)

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Output modality (text prose) matches the deployer's text-only acceptance [elicitation A4]. Both benchmark and deployment operate on Amharic text in Ethiopic script [Q79, Q382].

OF-2: Text-to-speech is not in scope for the benchmark or this deployment, but oral relay through cooperative leaders is a documented end-use pattern [regional context]; the benchmark does not evaluate any output that supports oral-relay quality (e.g., terminology consistency across read-aloud clauses).

OF-3: National adult literacy ~51.8% [WEB-5], with rural Amhara expected lower — translated prose will frequently be read aloud by cooperative leaders to lower-literacy borrowers, which depends on consistent administrative vocabulary not assessed by sentence-level chrF++ [Q455, Q648]. This is a partial form-mismatch insofar as the benchmark's evaluation unit (sentence) does not reflect the deployment's accessibility-mediated use unit (document read aloud).

OF-4: Form mismatch at the modality level is minimal; the principal residual gap is the absence of document-level/terminology-consistency scoring [Q5, Q336], which constrains how reliably aggregate scores predict fitness for binding documents.

ID	Evidence
Q5	"we evaluated the performance of over 40,000 different translation directions using a human-translated benchmark, Flores-200, and combined human evaluation with a novel toxicity benchmark covering all languages in Flores-200 to assess translation safety." (p.1)
Q79	"We represent each language with a BCP 47 tag sequence using a three-letter ISO 639-3 code as the base subtag, which we complement with ISO 15924 script subtags, as we collected resources for several languages in more than one script." (p.17)
Q336	"We look for best performance on the Flores-200 development set, and report detokenized BLEU on Flores-200 devtest." (p.41)
Q382	"We then trained an encoder for Amharic and Tigrinya only, paired with English as in all our experiments, and a specific 8k SentencePiece vocabulary to better support the Ge'ez script." (p.45)
Q455	"We use the chrF++ metric to compare the model performance (see Section 7.1)." (p.53)
Q648	"In this work, we primarily evaluate using chrF++ using the settings from sacrebleu. However, when comparing to other published works, we utilize BLEU and spBLEU where appropriate." (p.73)
WEB-5	Ethiopia adult literacy ~51.8% (UNESCO via World Bank, 2017) — establishes oral-relay relevance for output use (https://data.worldbank.org/indicator/SE.ADT.LITR.ZS?locations=ET)

Information Gaps

- Amhara-specific rural literacy rate not publicly disaggregated [registry NOT FOUND].

Remediation

Gap 1: All evaluation is sentence-level [Q5, Q336]; no document-level coherence or terminology-consistency scoring, which matters for oral-relay use to lower-literacy borrowers [WEB-5].

Recommendation: Add document-level evaluation: score multi-clause document translations on (a) cross-clause terminology consistency for defined terms, (b) preservation of numbered-clause structure in prose form, and (c) intelligibility when read aloud (oral-relay test with cooperative extension worker raters).

ID	Evidence
Q5	"we evaluated the performance of over 40,000 different translation directions using a human-translated benchmark, Flores-200, and combined human evaluation with a novel toxicity benchmark covering all languages in Flores-200 to assess translation safety." (p.1)
Q336	"We look for best performance on the Flores-200 development set, and report detokenized BLEU on Flores-200 devtest." (p.41)
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